# Research Statement

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## Research Vision

My research goal is to develop Vision-Language-Action (VLA) models that reason efficiently and generalize reliably under real-world constraints, including limited computation, delayed inference, and distribution shifts.

## Research Experience

My research began in September 2024 under **Prof. Tao He** at **Sichuan University**, focusing on ODE based medical image segmentation. I identified a trade off: discretized neural memory ODE decoders were efficient but numerically approximate, whereas continuous ODEs offered precise feature evolution at much higher cost and with limited cross domain generalization, which prompts me to investigate whether continuous ODE modeling could be made both compact and robust.

My first authored **AAAI 2026** paper, **CNM-UNet**, introduced a single Continuous Neural Memory ODE block that replaced multiple hierarchical decoder layers, achieving better segmentation accuracy while maintaining efficiency. I also designed a dual self updating strategy to improve cross domain adaptation. I led the project from problem formulation through experimentation and manuscript writing, with guidance from **Prof. He**.

The generalization challenges I encountered in **CNM-UNet** naturally led me to delve deeper into test-time adaptation. With **Prof. Tao He**, I explored continual adaptation, selective parameter updating, and optimizer-state reset, resulting in a paper under review at **Pattern Recognition** and another in preparation for **ICLR 2027**.

My presentation at **AAAI 2026** in Singapore exposed me to recent advances in embodied AI and multimodal reasoning, sparking my interest in how foundation models can reason and act in the physical world. In March 2026, I joined **Prof. Yuzhang Shang**'s group at the **University of Central Florida** as a remote research intern. I am currently investigating chain-of-thought reasoning and self-supervised reinforcement learning for VLA models. This experience has given me a grounding in the field and established efficient reasoning and deployment-time generalization as my primary research interests.

## Proposed Research

### Efficient Multimodal Reasoning for VLA

Explicit autoregressive CoT requires sequential token generation before action generation, substantially increasing inference latency and limiting its applicability in real-time closed-loop control. I therefore aim to replace explicit autoregressive reasoning with latent CoT that can be generated in a single forward pass, and develop lightweight supervision to enable efficient reasoning within compact latent representations.

Recent studies and my ongoing experiments suggest that multi-modal (visual and linguistic) CoT provides task-dependent and partially overlapping benefits. While each modality individually improves VLA performance, their joint improvement is often sub-additive, indicating that simply combining multiple CoT streams does not guarantee effective multimodal reasoning. I thus plan to investigate how multi-modal CoT can be integrated more effectively to leverage their complementary information for better action generation.

### Test-Time Self-Improvement for VLA

VLA models are typically fine-tuned on limited offline demonstrations, which can overfit the policy to the training distribution. However, real-world deployment inevitably encounters unseen objects, viewpoints, tasks, and physical conditions, leading to substantial performance degradation in novel environments.

I aim to develop unsupervised test-time adaptation methods that enable VLA models to improve through deployment-time interaction without additional human annotations. I plan to explore this direction under moderate distribution shifts within established benchmarks and more challenging sim-to-real transfer settings. Ultimately, I hope to enable VLA policies to continually adapt and self-improve when encountering previously unseen conditions.

## References

[1] Xu et al. "CNM-UNet: Continuous Ordinary Differential Equations for Medical Image Segmentation." AAAI, 2026.